GGG-R-400D
June 16, 1987
SUPERSEDING
GGG-R-400C
January 21, 1965

FEDERAL SPECIFICATION

RIVET SET, HAND

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand rivet sets to be used by blacksmiths', coppersmiths, tinsmiths, and saddlers in the setting of rivets.

1.2 Classification.

1.2.1 Types and style. Hand rivet sets shall be of the following types and style as specified (see 6.1).

Type I - Blacksmiths'
  Style A - Without handle

Type II - Coppersmiths', tinsmiths', and saddlers'

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Singles copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; Los Angeles; San Francisco; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their Agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society For Testing And Materials (ASTM) Standards:

ASTM E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Applications for copies should be addressed to the American Society For Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. Material used in the manufacture of hand rivet sets shall be forged tool steel properly heat-treated to produce tools conforming to the requirements specified herein. All forgings shall be of uniform quality, free from burrs, flash marks, or other defects or imperfections that may affect their serviceability and durability. Strength and other essential physical properties of the forgings shall be adequate throughout for the purpose intended (see 4.4).

3.2 Finish. All surfaces of the rivet sets shall be free from nodules, burrs, cracks, excessive oxide scale, and other detrimental defects. The rivet head end and the anvil end shall have a smooth finish, and all unfinished metal surfaces shall be either lacquered, enameled, painted, or have a natural finish with an oil coating.

3.3 Identification marking. The rivet sets shall be stamped or etched in a legible manner with the rivet size, manufacturer's name or a trademark and country of origin of such known character that the source of manufacturer may be readily identifiable.

3.4 Hardness.

3.4.1 Rivet-head end. The rivet-head end of the tool shall show a hardness of not less than 45 nor more than 55 on a Rockwell C scale.

3.4.2 Striking end. The striking end for type I shall show a hardness of not less than 32 nor more than 42 on Rockwell C scale.

3.5 Type I, blacksmiths'.

3.5.1 Style A, without handle. Style A rivet set shall be of a one piece construction and circular in cross section. One end of the rivet set shall be provided with a rivet set impression applicable to the size specified in tables I and II. The opposite end of the striking end of the rivet set shall be hemispherically rounded.

3.5.1.1 Rivet set sizes 1/4, 3/8, and 1/2 inch. The rivet-head end shall be larger than the shank for rivet set sizes 1/4, 3/8, and 1/2 inch. The sizes shall conform to table I as specified (see 6.1) and shall be similar to figure 1.

TABLE I. Type I, style A, blacksmiths', without handle, sizes 1/4, 3/8, & 1/2 inch

Rivet Size	Diameter of body A		Rivet set impression			Length of riveting head E		Overall length F		Length of taper (anvil end) G		Diam. of anvil end H
			Diam. B	Radius C	Depth D							
	Min	Max	Max	Max	Max	Min	Max	Min	Max	Min	Max	
Inch	Inches	Inches	Inch	Inch	Inch	Inch	Inch	Inches	Inches	Inch	Inch	Inch
1/4	11/16	13/16	7/16	7/32	3/16	13/16	15/16	5-3/8	5-5/8	3/4	7/8	5/16
3/8	15/16	1-1/16	21/32	21/64	9/32	13/16	15/16	5-7/8	6-1/8	3/4	7/8	9/16
1/2	1-1/4	1-3/8	29/32	31/64	5/16	13/16	15/16	5-7/8	6-1/8	3/4	7/8	7/8

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TABLE II. Type I, style A, blacksmiths', without handle, sizes 5/8 & 3/4 inch

Rivet Size	Diameter of body A		Rivet set impression			Overall length F		Length of taper (anvil end) G		Diam. of anvil end H
			Diam. B	Radius C	Depth D					
	Min	Max	Max	Max	Max	Min	Max	Min	Max	
Inch	Inches	Inches	Inches	Inch	Inch	Inches	Inches	Inch	Inch	Inch
5/8	1-5/16	1-7/16	1-1/8	19/32	13/32	5-3/4	6	11/16	13/16	5/8
3/4	1-7/16	1-9/16	1-11/32	23/32	31/64	6-5/32	6-13/32	11/16	13/16	5/8

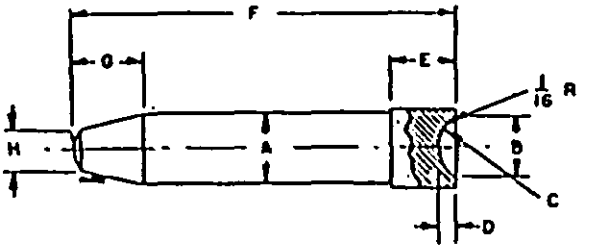


FIGURE 1. Type I, style A, blacksmiths' rivet set, without handle, sizes 1/4, 3/8 and 1/2 inch.

3.5.1.2 Rivet sizes 5/8 and 3/4 inch. Rivet sets for rivet sizes 5/8 and 3/4 inch shall have a straight shank, shall conform to table II for the size specified (see 6.1), and shall be similar to figure 2.

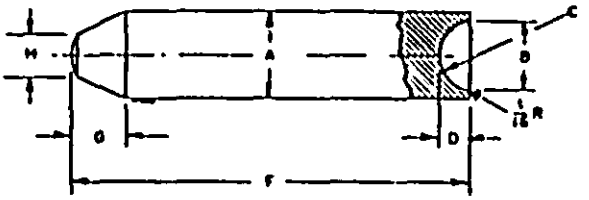


FIGURE 2. Type I, style A, blacksmiths', without handle, sizes 5/8 and 3/4 inch

3.6 Type II, coppersmiths', tinsmiths', and saddlers'. Type II rivet sets shall be either a rectangular cross section at the rivet head end or rectangular with short sides convex. The shank of the rivet set shall be either straight or tapered towards the striking end. All edges except the rivet head end of the tool shall be either chamfered or radiused. The rivet head end of the tool shall be provided with a rivet-set impression either angular or spherical and a rivet-set hole at right angle to the face of the tool. The rivet-set hole shall not be less than 3/8 inch deep. The face of the rivet head end shall be at right angle to the axis of the tool. A clearance hole shall be provided at the bottom of the rivet-set either from the short or long side of the tool. Type II rivet sets shall conform to table IV and shall be similar to figure 3.

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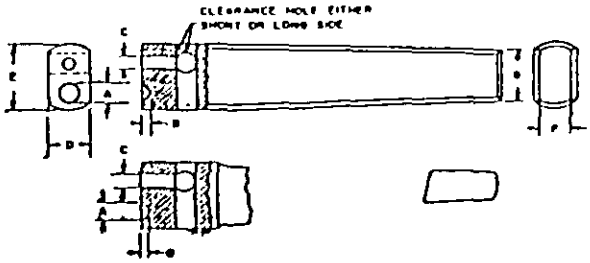


FIGURE 3. Type II, coppersmiths', tinsmiths' and saddlers' rivet sets.

TABLE III. Type II, coppersmiths', tinsmiths' and saddlers', rivet set

Size No.	Overall length		Rivet set impression Diameter A		Depth B		C	D		E		F^1	G^1	Suitable for Iron rivets Pounds per 1000	Copper rivets
			Angular shape	Spherical shape	Angular shape	Spherical shape									
	Min Inches	Max Inches	Max Inch	Max Inch	Max Inch	Max Inch	Min Inch	Min Inch	Max Inch	Min Inch	Max Inches	Min Inch	Min Inch		No.
00	5-1/4	5-1/2	33/64	33/64	0.149	0.125	0.3125	9/16	11/16	1	1-1/4	1/2	13/16	14	~
0	5-1/4	5-1/2	15/32	15/32	.135	.125	.265	9/16	11/16	1	1-1/4	1/2	13/16	10 & 12	5
1	4-1/2	5-1/4	13/32	7/16	.117	.093	.234	9/16	5/8	15/16	1-1/4	7/16	11/16	8	6
2	4-1/2	5-1/4	3/8	13/32	.108	.093	.213	1/2	5/8	15/16	1-1/16	7/16	11/16	6	7
3	4-1/2	4-13/16	11/32	21/64	.099	.078	.191	1/2	5/8	3/4	1-1/16	7/16	11/16	4 & 5	8
4	4-3/8	4-1/2	5/16	21/64	.090	.078	.166	1/2	19/32	3/4	1	3/8	9/16	2-1/2 & 3	9
5	4-3/8	4-1/2	9/32	21/64	.080	.062	.149	3/8	19/32	3/4	15/16	3/8	9/16	1-3/4 & 2	10&11
6	4-3/8	4-1/2	15/64	19/64	.068	.062	.140	3/8	19/32	3/4	15/16	3/8	9/16	1-1/2	12
7	4	4-1/2	7/32	17/64	.063	.046	.125	11/32	1/2	5/8	3/4	5/16	13/32	1-1/4	13
8	4	4-1/2	3/16	7/32	.054	.046	.109	11/32	1/2	5/8	3/4	5/16	13/32	5/8 & 3/4	14

1/ "F" and "G" dimensions applicable to tapered rivet sets (see 3.6).

3.7 Workmanship. Workmanship shall be of the highest grade throughout and equal in every respect to good commercial practice. The rivet sets shall be free from rust, burrs, fins, sharp edges, fractures, splits, malformed, broken or any imperfection which may impair their serviceability or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that the supplies and services conform to prescribed requirements.

4.4 Sampling for examination and test. Sampling for examination, inspection, and performance test shall be performed using the procedures prescribed by MIL-STD-105.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The sample unit for this examination shall be one completely fabricated rivet set. The inspection level shall be S-2 with an acceptable quality level (AQL) of 2.5 percent defective.

4.3.1.1 Examination of the end item for visual defects and workmanship. The rivet sets shall be visually examined for defects as outlined and classified in table IV.

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TABLE IV. CLASSIFICATION OF DEFECTS

Examine	Defect	Major	Minor
Finish	Missing or not as specified.	X	
	Nodules, burrs, cracks, oxide scale.		X
	Area of thin film or no film.		X
	Blistered, peeled, chipped, scratched, or foreign matter imbedded.		X
	Rivet head end and anvil end not smooth finish.		X
Design	Any characteristic not in accordance with the specified requirement.	X	
Construction and workmanship	Evidence of rust or corrosion.	X	
	Sharp burr or sharp edges that may be injurious to personnel.	X	
	Fractured, split, bowed, malformed, broken, or otherwise impaired.	X	
Marking identification	Omitted, missing, incomplete, not legible, or not as specified.		X

4.3.1.2 Examination for dimensional defects. Examination shall be made of the end item to determine compliance with dimensional requirements specified herein. Defects in dimensions shall be smaller or larger than dimensions specified.

4.3.1.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of section 5. The sample unit shall be one container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

4.3.2 Testing of the end item. Each sample unit shall be tested for performance as specified in 4.4 to determine compliance with the requirements specified therein. The inspection level shall be S-4 with an acceptable quality level (AQL) of 4.0 percent defective. In addition, each sample unit shall be tested for Rockwell C hardness as specified in 4.5 to determine compliance of 3.4.1 and 3.4.2. The inspection level shall be S-4 with an acceptable quality level of (AQL) of 4.0 percent defective in accordance with MIL-STD-105.

4.4 Performance test. Sample rivet sets shall be tested by heading at least 12 cold rivets of iron or steel. The rivets shall be of the size specified for that particular size tool and which would normally be headed with the size rivet set under test. For type I, style A rivet sets, the hammer used shall weigh not less than 32 ounces; and for type II the hammer shall weigh not less than 16 ounces. The sample rivet sets under test shall not bend, break, fracture, or otherwise show any indication of being made of unsuitable material or be improperly hardened and tempered.

4.5 Hardness test. Each sample unit shall be tested for Rockwell C hardness in accordance with ASTM E 18 to determine compliance with paragraphs 3.4.1 and 3.4.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or commercial, as specified (see 6.1).

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5.1.1 Level A. The rivet sets shall be cleaned, preserved, and packaged in accordance with level A of PPP-P-40.

5.1.2 Commercial. The rivet sets shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the receiving activity.

5.2 Packing. Packing shall be level A or commercial as specified (see 6.1).

5.2.1 Level A. The rivet sets shall be packed in accordance with level A of PPP-P-40. When specified (see 6.1), fiberboard boxes shall not be used for exterior shipping containers for domestic shipment.

5.2.2 Commercial. The rivet sets shall be packed in fiberboard boxes to insure delivery at destination, to provide for redistribution by the initial receiving activity, and shall be acceptable by common carrier under the National Motor Freight Classification and the Uniform Freight Classification.

5.3 Marking. Marking shall be as specified in the contract or order.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number and date of this specification.
(b) Type and style (if applicable) of rivet sets required (see 1.2).
(c) Size number required (see applicable tables).
(d) Level of preservation, packaging and packing (see 5.1 and 5.2).
(e) When fiberboard boxes shall not be used for domestic shipment (see 5.2.1).

PREPARING ACTIVITY

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